TRANSITION ALGORITHM SPECIFICATION

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0.1. Bounded RC graphs.

Definition 0.1.1. Let $R \subseteq \mathbb{P} \times \mathbb{P}$ be a finite set. To each such set we associate an element w_R of S_∞ as follows. Given a pair (i, j) where $i, j > 0$ are integers, define $s(i, j) = s_{i+j-1}$. Totally order the grid such that $(i, j) \prec (a, b)$ if and only if $i < a$ or $i = a$ and $j > b$ (in other words, lexicographical order, except that the order on the second coordinate is reversed). By this ordering, index R as $r_1, r_2, \ldots, r_m$ in increasing order. Then

$$w_R = s_{r_1} \cdots s_{r_m}$$

If $\ell(w_R) = m$, then we say that R is an *RC-graph.*

Define $\mathfrak{m}(w)$ to be the index of the largest descent of a permutation w. A *bounded RC-graph* is an RC-graph R together with a specified number of rows $n \geq \max\{i : (i, j) \in R \text{ for some } j\}$, with the added condition that $\mathfrak{m}(w_R) \leq n$. The definition is as an ordered pair (R, n), but it will be far more convenient to abuse notation and consider the number of rows a property of R itself. To denote the number of rows, we define the *height* of R to be $\mathtt{ht}(R) = n$. A bounded RC graph has an associated weight vector $\mathtt{wt}(R)$ where $\mathtt{wt}_i(R)$ is the number of elements in row i.

The *word* of R, denoted $\mathrm{word}(R)$, is the sequence of indices $i + j - 1$ of the simple reflections $s_{r_1}, \ldots, s_{r_m}$ read in the order above; it is a reduced word for w_R. For a pair of positive integers (i, j) we define the *root*

$$\mathtt{rt}_R(i, j) = (w_{R[i,j]}^{-1}(i + j - 1), w_{R[i,j]}^{-1}(i + j)), \qquad R[i, j] = \{(a, b) \in R \mid (a, b) \succ (i, j)\}.$$

(this is the right root of the letter at the same position in the word).

We write $\mathcal{RC}_n$ for the set of bounded RC graphs of height n, $\mathcal{RC}_n(w)$ for those with permutation w, and $\mathcal{RC}_n(w)^0$ (or $\mathcal{RC}_n^0$) for those whose last row is empty.

0.2. The transition map and clipping.

We now construct a map

$$\mathcal{Z} : \mathcal{RC}_n^0 \to \mathcal{RC}_{n-1}$$

on bounded RC graphs with empty last row, realizing on the level of RC graphs the Lascoux–Schützenberger transition formula for Schubert polynomials (see [2]). The map is an iteration of Little bumps [1], realized on RC graphs by an explicit insertion algorithm; working with the insertion description keeps the compatible-sequence structure in view at every step, so that the output is manifestly again a bounded RC graph.

Algorithm 1 Basic monk insertion $i \overset{k}{\rightsquigarrow} R$

1: **Input:** An RC graph R, parameter k, and an integer i with $1 \leq i \leq k$.
2: **Output:** A modified RC graph R' with $\mathtt{wt}_j(R') = \mathtt{wt}_j(R)$ for $j \neq i$ and $\mathtt{wt}_i(R') = \mathtt{wt}_i(R) + 1$ such that $w_R \xrightarrow{k} w_{R'}$.
3: Find leftmost position $(i, j) \notin R$ in row i such that if $\mathtt{rt}_R(i, j) = (a, b)$, then $a \leq k < b$.
4: $R \leftarrow R \cup \{(i, j)\}$
5: **if** there exists $(i', j') \in R$ with $\mathtt{rt}_R(i', j') \in \Phi^-$ **then**
6: $\quad R \leftarrow R \setminus \{(i', j')\}$
7: $\quad$ Return to step 3 substituting $i = i'$.
8: **end if**
9: **return** R

Here $u \xrightarrow{k} u'$ means $u' = ut_{ab}$ for a single transposition with $a \leq k < b$ and $\ell(u') = \ell(u) + 1$, and Φ^- denotes the set of negative roots, i.e. pairs (a, b) with $a > b$.

Lemma 0.2.1 (Insertion validity)**.** *Algorithm 1 terminates, and its output R' is a valid RC graph with*

$$\mathtt{wt}_j(R') = \mathtt{wt}_j(R)\ (j \neq i), \qquad \mathtt{wt}_i(R') = \mathtt{wt}_i(R) + 1, \qquad w_R \xrightarrow{k} w_{R'}.$$

Proof. The first step adds one crossing in row i: adding the crossing (i,j) found in step 3 adjoins the root $t_a - t_b$, $a \leq k < b$, to the inversion set, so the permutation is multiplied by t_{ab} and the length rises by one, unless the addition creates a negative root elsewhere. A negative root can only have been created by the added crossing, and by the strong exchange property it is unique; moreover it occurs at a crossing (i', j') with $i' < i$. Indeed, the root $\mathtt{rt}_R(i', j')$ of a crossing is computed from the crossings following it in the total order of Definition 0.1.1, so it is unchanged unless (i', j') precedes (i, j), that is, $i' < i$ or $i' = i$ with $j' > j$; and within row i the crossings at columns $j' > j$ retain their roots as well, since the reflections contributed by row i to the right of column j' are composed in the same relative order with (i, j) inserted after them. Every subsequent step removes the crossing carrying the negative root and immediately replaces it in the row it was initially in: the removal restores the permutation to its value at the start of the pass, and the algorithm re-enters step 3 in that same row. Since the rectification row strictly decreases at every occurrence, the cascade terminates, and the final pass performs a clean addition, multiplying the permutation by a single transposition t_{ab} with $a \leq k < b$. The weight bookkeeping is therefore immediate: the first step contributes one crossing to row i, and each rectification removes and replaces one crossing within a single row, so $\mathtt{wt}_i(R') = \mathtt{wt}_i(R) + 1$ and all other row weights are unchanged. □

Definition 0.2.2 (Little bumps [1])**.** Given a reduced word $a_1 \cdots a_m$ for a permutation w and an index $1 \leq p \leq m$, the *Little bump* $(a_1 \cdots a_m){\uparrow_p}$ is defined recursively as follows. If $a_p > 1$, replace a_p with $a_p - 1$, and if $a_p = 1$ then instead replace all a_q with $q \neq p$ with $a_q + 1$. Denote the new word by $a'_1 \cdots a'_m$. If this resulting word is reduced, we are done. Otherwise, there is a unique index $j \neq p$ such that the word obtained by deleting a'_j from the new word is reduced by the deletion property of Coxeter groups. We then define

$$(a_1 \cdots a_m){\uparrow_p} = (a'_1 \cdots a'_m){\uparrow_j}$$

Definition 0.2.3 (Single-bump map)**.** Let R be a bounded RC graph with $\mathtt{ht}(R) = m$, empty last row, and $\mathfrak{m}(w_R) = m$. By the exchange property there is a unique crossing $(i, j) \in R$ with $\mathtt{rt}_R(i, j) = (m, m+1)$, and $i < m$. Set

$$R' = (R \setminus \{(i, j)\}) \cup \{(m, 1)\},$$

and define

$$\mathscr{L}(R) = (i \overset{m-1}{\rightsquigarrow} R') \setminus \{(m, 1)\},$$

regarded as a bounded RC graph of height $\mathfrak{m}(w_{\mathscr{L}(R)})$; all its crossings lie in rows $1, \ldots, m-1$.

Proposition 0.2.4 (The bump map is a Little bump)**.** *Let R be as in Definition 0.2.3 with $m = \mathfrak{m}(w_R)$, and let p be the index of the right root $(m, m+1)$ in* $\mathrm{word}(R)$*. Then*

$$\mathrm{word}(\mathscr{L}(R)) = \mathrm{word}(R){\uparrow_p}\ .$$

Proof. FILL IN PROOF HERE □

References

[1] Little, D. P. Combinatorial aspects of the Lascoux–Schützenberger tree. *Advances in Mathematics 174*, 2 (2003), 236–253.

[2] Macdonald, I. G. *Notes on Schubert Polynomials.* Université Du Québec à Montréal, Dép. de mathématiques et d'informatique, 1991.